Frayer models

SECTION A / VOCABULARY Name _____

Find the word on the screen. Talk it through with your partner, then write or sketch in its four boxes.

Evidence

Meaning in my words How would you explain evidence to a friend?	**How can you tell?** What reading supports the claim that the cart reached 4 m in 2 s?
Example or sketch How could two measurements support a claim that one cart accelerated more?	**Non-example and why** “I expected our cart to win.” Does that tell us which cart crossed first?

Inference

Meaning in my words How would you explain inference to a friend?	**How can you tell?** Wet tracks run from the door to a cart. Which part is your inference?
Example or sketch Two timed images show XJ closer to Earth. What can you reasonably infer?	**Non-example and why** “The second image reads 4 km.” Is that your inference or your reading?

Frayer models

SECTION A / VOCABULARY Name _____

Find the word on the screen. Talk it through with your partner, then write or sketch in its four boxes.

Model

Meaning in my words How would you explain model to a friend?	**How can you tell?** A cart moves 2 m each second. What are we assuming if we extend that pattern?
Example or sketch Where would your model put the cart one second after it passes 6 m going right?	**Non-example and why** The cart hits a cushion. Would you still extend the same pattern? Why?

Testable question

Meaning in my words How would you explain testable question to a friend?	**How can you tell?** What would you measure to answer “Does this cart speed up on the ramp?”
Example or sketch What question could two timed pictures of the same cart help you answer?	**Non-example and why** “Which cart looks coolest?” What would you change so the class can test it?

Doodle the idea

SECTION B / DRAW WITH YOUR TEACHER Name _____

Draw a three-sighting timeline. Label the starting position, change per second, and one future position. Use a dotted mark for the forecast.

Choose zero, direction, and scale.



My marks mean

One more exposure

01

SECTION E FINAL MEETING / TEAM TRANSFER

Name _____ Partner _____

Our goal: Separate a sighting from a prediction.

CHOOSE ONE JOB TOGETHER

TELESCOPE SCHEDULER ☐

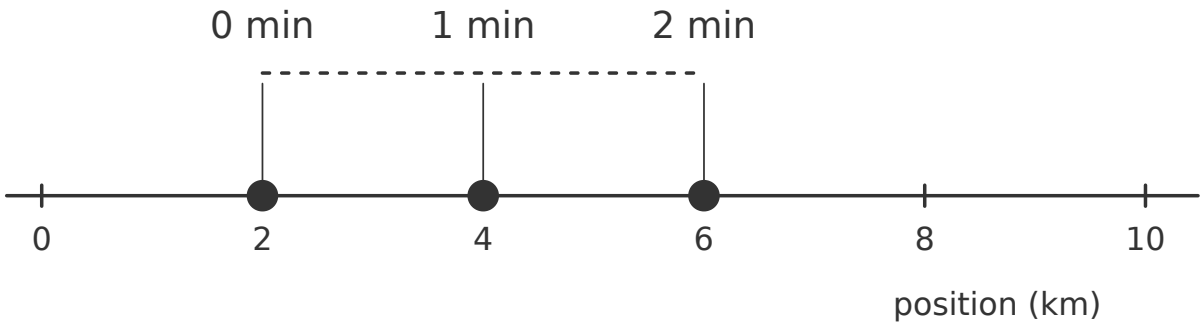
Mark where to aim at minute **3**. Tell the operator which dots were measured and which dot you predicted.

SCIENCE BROADCASTER ☐

Give viewers a two-sentence update. Point to a measured dot, then say where you expect the next one.

Point to a clue. Keep your first marks; add the update using the color key.

Three telescope fixes are at **2**, **4** and **6 km** at minutes **0**, **1** and **2**. The night crew has one more exposure.



01 OUR FIRST CALL

Where should the camera look next? Use the timed dots in your call.

02 OUR REVISED CALL

Wait for the reveal. Keep your first call.

New clue: _____

Does the new sighting fit your prediction? Tell the next crew what to check.

Switch speakers. What new observation would make you change your prediction? Explain why.

☐ GREEN	☐ RED	☐ YELLOW	☐ BLUE	☐ PURPLE	☐ ORANGE	☐ PENCIL
speeding up	slowing down	steady / balanced	position / displacement	velocity	forces	other marks